SmartAPI – Integration Specifications

Version 2.10

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TABLE OF CONTENTS

Table of Contents 3
SmartAPI Overview 5
How to Use this Document 5
API Connectivity 6
HTTP Post URL 6
Authentication Headers 6
Encryption Protocols 6
Verification Service Details 7
Introduction 7
Automated Valuation Model (AVM) 8
Business Credit Report 10
Consumer Credit Report 11
Criminal Record Report 14
Eviction Report 15
Flood Zone Determination 16
InstantID 17
Mortgage Electronic Registration Systems (MERS) 18
Mortgage Participant Verification (MPV) 19
Packaged Report 20
Precise ID Knowledge IQ 22
Precise ID Score 23
SSA-89 24
Tax Transcript (TAX) 25
Undisclosed Debt Notification (UDN) 27
Verification of Asset (VOA) 32
Verification of Deposit (VOD) 34
Verification of Employment (VOE) 35
Appendix 38
Duplicate Order Detection 38
Error Checking 39
Enumerations 41
InterfaceToken 41
Namespaces & Schemas 42

Response Formats 43
Surrogate Login 44
Test Cases 45

SMARTAPI OVERVIEW

MeridianLink's SmartAPI is a single XML interface to order verification services from service providers who leverage MortgageCreditLink™ (MCL) fulfillment technology. With SmartAPI, integrators can send a single XML request and the fulfillment technology will respond with the verification services appropriate to the request.

SmartAPI supported verification services include, but are not limited to:

- **Credit Verification** – Consumer Credit Reports and Undisclosed Debt Notifications from the national credit reporting agencies (Equifax, TransUnion, and Experian), credit analysis tools from CreditXpert™, and Business Credit Reports.
- **Property Verification** - Flood Certifications and Automated Valuation Models.
- **Employment Verification** – Instant verifications from The Work Number® and manual verifications of employment
- **Income Verification** – Tax return transcript verification directly from the IRS.
- **Asset and Deposit Verification** – Electronic verification of assets with AccountChek™ and manual verification of deposit.
- **Identity Verification, Validation, and Authentication** – Red Flags Identity Reports, SSA-89 SSN Validation, and Knowledge-based/Out-of-Wallet authentication
- **Watchlist Verification** – OFAC, SAM, National Appraiser Registry, Customer Uploaded Lists and others to assist with compliance in BSA, USA Patriot Act, and Fannie Mae LQI.
- **Packaged Reports** – Customizable fraud detection reports for data verification, validation, and discovery on borrowers, properties, and participants.

HOW TO USE THIS DOCUMENT

This document is comprised of several sections:

The **API Connectivity** section describes network level requirements, authentication, and connectivity recommendations for SmartAPI.

The **Verification Service Details** section contains service-specific workflows and additional information relevant to each service.

The **Appendix** includes additional miscellaneous information, such as test consumers and error handling guidance.

API CONNECTIVITY

HTTP POST URL

SmartAPI XML requests will be sent via HTTP POST to a specific URL that will vary depending on your service provider. Each service provider on the MCL platform will have their own unique URL. The test server is the one that you will be working with and its URL is below:

https://demo.mortgagecreditlink.com/inetapi/request_products.aspx

AUTHENTICATION HEADERS

Authentication is handled via HTTP headers. Login credentials and interface identifier must be included in the headers.

LOGIN CREDENTIALS

Each request will require a user's login name and corresponding password for authentication. In a production environment, each end-user will typically be assigned their own login/password by their service provider. In special cases, a company or branch will have a single set of credentials shared by all users. Speak with your technical representative if you are unsure of which approach to support.

A set of test credentials will be provided to you during initial communication with your MeridianLink technical representative. The login and password should be sent via an "Authorization" header using the "Basic" schema.

For example, if your login is "john78" and your password is "home123", then the Base64 encoding of "john78:home123" is "am9objc4OmhvbWUxMjM=". What you would send as part of the header's value:

Authorization: Basic am9objc4OmhvbWUxMjM=

INTERFACE IDENTIFIER

This HTTP header value is used to identify your system, its access rights, and other configuration settings. During the development phase, you will use a test value. This test identifier will only work when sending requests to the test URL. Once you are ready for production, you will be given a new identifier unique to your system that you will use in place of this test identifier.

The header field to use is "MCL-Interface". For testing, please use a value of "SmartAPITestingIdentifier".

MCL-Interface: SmartAPITestingIdentifier

Note: Production interface identifiers should be kept confidential and not shared between different software systems. For example, if you plan on building a white label product, then each copy of the software sold should utilize a unique interface identifier (which can be obtained by reaching out to us). Sharing the same interface identifier between different systems will result in configuration changes intended for one system to also apply other systems using that same identifier.

ENCRYPTION PROTOCOLS

The communication channel uses a secured HTTP post. The encryption protocol must be TLS 1.2 or higher. If your application receives no response from our endpoint, please check that you are using TLS 1.2 or higher and a compatible cipher suite. Supported cipher suites can be determined by using various free online tools.

VERIFICATION SERVICE DETAILS

INTRODUCTION

BASIC WORKFLOW

When a new order is submitted to the SmartAPI interface, a unique file number (i.e. VendorOrderIdentifier) will be generated and echoed in the response. The file will then flow through 2 status phases: a “processing” phase and an “end” phase. A “processing” phase (e.g. New, Processing, Pending, etc) means the order is in progress. An “end” phase means the file is done (e.g. Completed. Canceled, Error, etc). When you initially submit your new order, you should poll/query for the status of the file every one to ten seconds. Eventually, you will receive a status indicating the order is at an “end” phase, which means you can stop polling. If the end status was a successful completion (as opposed to an error), then your completed report will also be included in the response.

Each product’s workflow may vary slightly from what was described above and such differences will be mentioned, but overall, this is how services are fulfilled with the SmartAPI interface.

XML SAMPLES

The kit contains sample XML request and response files to help you during your integration. The response files are samples of what you can expect to receive on completed orders. The request files follow a certain naming convention to help identify which samples model what type of request:

- **Order New**: These contain a sample of a new order request and use only the data points that are needed (i.e. required fields) to fulfil the service. These samples are ready to use and can be submitted to the SmartAPI interface as-is.
- **Retrieve Existing**: These are samples of what you would need to send on a polling/query request. The terms “poll”, “query”, and “retrieve” are often used interchangeably with regards to the SmartAPI interface. This is because such requests will not only echo back the status of your file (e.g. polling for the status; status query), but if the file is completed, will also include the data for it (e.g. report retrieval). This single action fulfills both use cases.
- **Advanced Sample**: These samples contain optional data points that can be included with your request, as well as important enumerations for certain elements. These samples are for reference purposes and are not designed to be submitted to the SmartAPI interface directly. Questions surrounding any nodes in this document should be directed to your technical representative.

Other types of samples may also be present, depending on the product, and we hope the names used for them are self-explanatory. If you have any questions, please speak with your technical representative.

AUTOMATED VALUATION MODEL (AVM)

SmartAPI supports the following automated valuation models: PASS, Home Value Explorer (HVE), IVAL, GeoAVM, and the Property and Market Condition (PMC) reports. Unlike the other models, PMC reports are manually processed and can take several days to complete.

WORKFLOW

When a new order is submitted (or a query of an existing order), the response XML will include a StatusCode value that appears under the //SERVICE/STATUSES/STATUS container. That StatusCode will dictate your required follow-up action. Below are the appropriate actions to take, based on the StatusCode returned:

- **New**: The system received your recent order request and is processing it. A unique VendorOrderIdentifier was generated for this order and you should begin polling for that until you receive an end status.
- **Processing**: The system is continuing to process your order. Continue to poll for the VendorOrderIdentifier until you receive an end status.
- **Completed**: The order is complete and the data you've requested is included in the response. This is considered an end status.
- **Error**: An error occurred while processing the order. A description should be included under the StatusDescription element. Please display this to the end-user. This is considered an end status.
- **NoHit**: The property could not be located or a valuation for it could not be generated. A StatusDescription might be present. If so, display it to the end-user. This is considered an end status.

POLLING GUIDELINES

The following polling rules should be adhered to:

- For GeoAVM, HVE, IVAL, and PASS: if the VendorOrderIdentifier is not in an end status after 90 seconds from the time the new order request was submitted, you should time out the order and instruct the end-user to reach out to their service provider. These orders should not take this long to process, so the system is likely experiencing an issue.
- For PMC: if the VendorOrderIdentifier is not in an end status after 30 seconds from the time the new order request was submitted, your order is being processed manually and it will take several days for completion. We recommend automatically polling for it once or twice a day to ensure the status of the order on your end is in sync with our system.
- We recommend allowing the user to manually submit a retrieval request on orders that are being manually processed. This will allow them to immediately retrieve the completed report once they are notified of its completion, rather than waiting for the next polling run.
- Only one AVM product should be ordered (i.e. flag set to "true") per submission, unless the Stop On First Hit flag is set to "true". In the case where Stop On First Hit is set to "true", multiple AVM products can be requested, but only one VendorOrderIdentifier will be created by our system.
- The PMC product cannot be ordered in conjunction with the other AVM products. If PMC and other AVM products are needed for a loan, PMC should be submitted as a separate request.

STOP ON FIRST HIT

The Stop on First Hit flag (included in the XML request) is intended to be used to initiate automatic cascade ordering. When set to true, it will instruct the system to order the indicated AVM models one at-a-time and stop when the first successful result is received.

The sequence in which the system orders the AVMs is based off the user's preference, which is configured by the user via our web portal.

CASCADING OPTIONS

There are 2 cascading approaches that are commonly used to order AVMs:

Manual Cascade

The user orders a single AVM model at a time. Generally, only a single, preferred AVM will be ordered by the user. If for some reason no data can be generated for that property, the user tends to follow-up by submitting an order for their next preferred AVM model, and so on. The workflow used here would be considered cascading, but it's all done manually or, at the least, not managed by the SmartAPI system.

Automated Cascade

Order multiple AVM models in a single request and set the Stop On First Hit flag to "true". This will carry out the same process in the manual cascade workflow, but the cascading is handled by SmartAPI. Our system will cascade through the selected models and stop at the first successful result. As mentioned in an earlier section, the AVM ordering preference is controlled by the end-user via our web portal. Below is an example of how this would work:

Let's say the user has configured their AVM ordering as follows, with the most preferred first:

1. HVE
2. IVAL
3. PASS
4. GeoAVM.

Let's say an order is submitted with the below XML flags:

RequestGEOAVM	true
RequestHVE	true
RequestIVAL	true
RequestPASS	false
RequestStopOnFirstHit	true

Our system will process this request by first trying HVE. If HVE does not return a successful result, we'll then try IVAL. If IVAL does not return a successful result, we'll then try GeoAVM. Notice that we skipped the PASS product because it was set to "false" in the request. With automated cascade, though multiple orders may be generated behind-the-scenes as we look for a successful result, only a single VendorOrderIdentifier is created.

BUSINESS CREDIT REPORT

SmartAPI supports business credit reports, which are similar to consumer credit reports, but are used to pull data on legal entities.

WORKFLOW

When a new order is submitted (or a query of an existing order), the response XML will include a StatusCode value that appears under the //SERVICE/STATUSES/STATUS container. That StatusCode will dictate your required follow-up action. Below are the appropriate actions to take, based on the StatusCode returned:

- **New**: The system received your recent order request and is processing it. A unique VendorOrderIdentifier was generated for this order and you should begin polling for that until you receive an end status.
- **Processing**: The system is continuing to process your order. Continue to poll for the VendorOrderIdentifier until you receive an end status.
- **Completed**: The order is complete and the data you've requested is included in the response. This is considered an end status.
- **Error**: An error occurred while processing the order. A description should be included under the StatusDescription element. Please display this to the end-user. This is considered an end status.

POLLING GUIDELINES

The following polling rules should be adhered to:

- If the VendorOrderIdentifier is not in an end status after 90 seconds from the time the new order request was submitted, you should time out the order and instruct the end-user to reach out to their service provider. These orders should not take this long to process, so the system is likely experiencing an issue.

CONSUMER CREDIT REPORT

SmartAPI supports ordering credit data from the three major bureaus: Equifax, Experian, and Transunion.

WORKFLOW

When a new order is submitted (or a query of an existing order), the response XML will include a StatusCode value that appears under the //SERVICE/STATUSES/STATUS container. That StatusCode will dictate your required follow-up action. Below are the appropriate actions to take, based on the StatusCode returned:

- **New**: The system received your recent order request and is processing it. A unique VendorOrderIdentifier was generated for this order and you should begin polling for that until you receive an end status.
- **Processing**: The system is continuing to process your order. Continue to poll for the VendorOrderIdentifier until you receive an end status.
- **Completed**: The order is complete and the data you've requested is included in the response. This is considered an end status.
- **Error**: An error occurred while processing the order. A description should be included under the StatusDescription element. Please display this to the end-user. This is considered an end status.

POLLING GUIDELINES

The following polling rules should be adhered to:

- If the VendorOrderIdentifier is not in an end status after 90 seconds from the time the new order request was submitted, you should time out the order and instruct the end-user to reach out to their service provider. These orders should not take this long to process, so the system is likely experiencing an issue.

CREDIT DATA, SCORE ADD-ON, AND FRAUD ADD-ON

MCL integrates to all 3 major credit bureaus: Equifax, Experian, and TransUnion. When requesting credit data, MCL also has the option to request add-on products, such as credit scores and fraud searches. Likewise, SmartAPI supports indicating these preferences with each order via use of flags. The flags would be inserted under the CREDIT_REPOSITORY_INCLUDED group within the SERVICE container. Please reference the XML samples included with this kit to gain an understanding of the layout. The following flags are available:

	Credit Data	Score Add-On	Fraud Add-On
Equifax	X	X	
Experian	X	X	X
TransUnion	X	X	X

Credit scores and fraud searches are add-on products; thus, they will only be ordered if credit data for the corresponding bureau is also being ordered. Otherwise, requests for them will be ignored. Each of the flags supports 3 states:

State	Description
true	This indicates a preference to include the corresponding product or add-on as part of the order.
false	This indicates a preference to not include the corresponding product or add-on as part of the order.
Omitted	This indicates a preference to let the service provider's configurations for the user's account determine how that item will be handled. The omission being referenced is to the element itself—it should be removed.

Note that we specifically use the word "preference" when describing the function of the flags. While the user can indicate their preferences when ordering, the configuration the service provider has on each user's account determines whether that preference is honored or ignored. The service provider can configure each option with the following granularity:

Configuration	Description
Disabled	The item will never be ordered, regardless of the user's indicated preference.
Optional	The item will not be ordered, unless a preference of 'true' is submitted for it.
Default	The item will be ordered, unless a preference of 'false' is submitted for it.
Always	The item will always be ordered, regardless of the indicated preference.

ORDERING A REFRESH REPORT

The refresh report is a soft inquiry ran on the applicant just prior to the close of the loan. This report is used to ensure the applicant has not taken on any new liabilities, which may trigger an adjustment to their loan terms or eligibility. As its name suggests, refresh reports cannot be ordered standalone; they must include a reference to the initial credit report that was ordered.

To order a refresh, include the file number of the initial credit pull (i.e. the hard inquiry). The system will respond with a new file number, which you should poll for as you would a new credit order.

ORDERING A PREQUAL SOFT INQUIRY

In SmartAPI, ordering a prequal report is the same workflow and XML schema as ordering a traditional consumer credit report. The subscriber codes and purpose codes are what will determine if a credit report is a hard pull or soft pull and this is configured at the account level for the user by the CRA. Typically, prequal soft pull accounts are separate from the regular hard pull accounts so users will utilize separate logins to differentiate between the two. Please contact your MeridianLink technical provider if you have any additional questions regarding prequal reports or would like to discuss further.

PERMANENT UNMERGING OF A REPORT

In this scenario, you have a credit report where you need to remove a borrower and/or bureau(s) from the existing data set. An example of where this would be used is if the user initially orders a joint credit report and determines that one of the applicant's scores are too low to qualify for a loan. Instead of repulling credit for the qualifying applicant to get a report with just that person, an unmerge can be done. The unmerge request must contain the existing credit file number from which the unmerge will apply, along with the borrower(s) and/or bureau(s) you wish to retain in the unmerged report.

This action will not make any changes to the original credit file; instead, a new file will be created to reflect the unmerged data.

Unmerging a file does take a few moments to process, so after an unmerge request is submitted, please begin polling for the new file number echoed in the response as you would with a new order request.

UPGRADING A REPORT

Upgrading a credit report means adding a coborrower and/or bureaus to an existing file. The upgrade request must contain the file number you intend to upgrade, the data that already exists on the file, and the additional borrower or bureaus that will be added. Upgrade requests can be sent to repull frozen/deceased bur infile(s) or upgrade the XP Power Profile soft pull if your account is enabled for access. Please reach out to your integration POC to configure your test account.

This action will not re-order any bureaus that were already successfully ordered on the file; SmartAPI will add only what is applicable.

Depending on the service provider's configuration, upgrading a file may result in a new file number (i.e. VendorOrderIdentifier) being created to hold the new data set. In such a scenario, that new VendorOrderIdentifier will be echoed in the initial response to the upgrade request.

Upgrading a file does take a few moments to process, so after an upgrade request is submitted, please begin polling for the VendorOrderIdentifier returned in the initial response.

DETERMINING IF A FILE IS A SOFT PULL OR HARD PULL

Within the XML response, look for the following which will indicate a soft pull:

<CreditReportType>Other</CreditReportType>

<CreditReportTypeOtherDescription>SoftPull</CreditReportTypeOtherDescription>

CRIMINAL RECORD REPORT

SmartAPI supports Criminal Record Reports through Contemporary Information Corp.

WORKFLOW

When a new order is submitted (or a query of an existing order), the response XML will include a StatusCode value that appears under the //SERVICE/STATUSES/STATUS container. That StatusCode will dictate your required follow-up action. Below are the appropriate actions to take, based on the StatusCode returned:

- **New**: The system received your recent order request and is processing it. A unique VendorOrderIdentifier was generated for this order and you should begin polling for that until you receive an end status.
- **Processing**: The system is continuing to process your order. Continue to poll for the VendorOrderIdentifier until you receive an end status.
- **Completed**: The order is complete and the data you've requested is included in the response. This is considered an end status.
- **Error**: An error occurred while processing the order. A description should be included under the StatusDescription element. Please display this to the end-user. This is considered an end status.

POLLING GUIDELINES

The following polling rules should be adhered to:

- If the VendorOrderIdentifier is not in an end status after 90 seconds from the time the new order request was submitted, you should time out the order and instruct the end-user to reach out to their service provider. These orders should not take this long to process, so the system is likely experiencing an issue.

EVICTION REPORT

SmartAPI supports Eviction Reports through Contemporary Information Corp.

WORKFLOW

When a new order is submitted (or a query of an existing order), the response XML will include a StatusCode value that appears under the //SERVICE/STATUSES/STATUS container. That StatusCode will dictate your required follow-up action. Below are the appropriate actions to take, based on the StatusCode returned:

- **New**: The system received your recent order request and is processing it. A unique VendorOrderIdentifier was generated for this order and you should begin polling for that until you receive an end status.
- **Processing**: The system is continuing to process your order. Continue to poll for the VendorOrderIdentifier until you receive an end status.
- **Completed**: The order is complete and the data you've requested is included in the response. This is considered an end status.
- **Error**: An error occurred while processing the order. A description should be included under the StatusDescription element. Please display this to the end-user. This is considered an end status.

POLLING GUIDELINES

The following polling rules should be adhered to:

- If the VendorOrderIdentifier is not in an end status after 90 seconds from the time the new order request was submitted, you should time out the order and instruct the end-user to reach out to their service provider. These orders should not take this long to process, so the system is likely experiencing an issue.

FLOOD ZONE DETERMINATION

SmartAPI supports flood determination orders through various providers.

WORKFLOW

When a new order is submitted (or a query of an existing order), the response XML will include a StatusCode value that appears under the //SERVICE/STATUSES/STATUS container. That StatusCode will dictate your required follow-up action. Below are the appropriate actions to take, based on the StatusCode returned:

- **New**: The system received your recent order request and is processing it. A unique VendorOrderIdentifier was generated for this order and you should begin polling for that until you receive an end status.
- **Processing**: The system is continuing to process your order. Continue to poll for the VendorOrderIdentifier until you receive an end status.
- **Pending**: Automatic determination is not possible for this property and so your request is being processed manually. A StatusDescription element may accompany this status as it is worked on.
- **Completed**: The order is complete and the data you've requested is included in the response. This is considered an end status.
- **Canceled**: The order was in Pending status and subsequently canceled, typically at the request of the end-user. This is considered an end status.
- **Error**: An error occurred while processing the order. A description should be included under the StatusDescription element. Please display this to the end-user. This is considered an end status.
- **Transferred**: The order was completed and then the user transferred the flood certificate to another party. From a technical perspective, this can be treated as a Completed status.

POLLING GUIDELINES

The following polling rules should be adhered to:

- If the VendorOrderIdentifier is in a New or Processing status for more than 90 seconds from the time the new order request was submitted, you should time out the order and instruct the end-user to reach out to their service provider. Flood determinations should either reach an end status or Pending status within 90 seconds; any longer and there is likely a system issue.
- Orders that are in Pending status will take several days to weeks to complete. Therefore, we recommend automatically polling for the order once or twice a day until it reaches an end status. This will ensure the status you have on your end is in sync with what we have and that your system will pull in the completed report when the order is done.
- We recommend allowing the user to manually submit a retrieval request on orders that are being manually processed. This will allow them to immediately retrieve the completed report once they are notified of its completion, rather than waiting for the next polling run.

INSTANTID

SmartAPI supports Lexis Nexis' InstantID product, which provides identity verification and risk assessment with regards to fraud and identity theft.

WORKFLOW

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- **New**: The system received your recent order request and is processing it. A unique VendorOrderIdentifier was generated for this order and you should begin polling for that until you receive an end status.
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- **Completed**: The order is complete and the data you've requested is included in the response. This is considered an end status.
- **Error**: An error occurred while processing the order. A description should be included under the StatusDescription element. Please display this to the end-user. This is considered an end status.

POLLING GUIDELINES

The following polling rules should be adhered to:

- If the VendorOrderIdentifier is not in an end status after 90 seconds from the time the new order request was submitted, you should time out the order and instruct the end-user to reach out to their service provider. These orders should not take this long to process, so the system is likely experiencing an issue.

MORTGAGE ELECTRONIC REGISTRATION SYSTEMS (MERS)

SmartAPI supports ordering of MERS reports, which provides a list of all mortgages associated with a consumer. Searches for subject properties can also be ran, but only in conjunction with a search for a consumer.

WORKFLOW

When a new order is submitted (or a query of an existing order), the response XML will include a StatusCode value that appears under the //SERVICE/STATUSES/STATUS container. That StatusCode will dictate your required follow-up action. Below are the appropriate actions to take, based on the StatusCode returned:

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MORTGAGE PARTICIPANT VERIFICATION (MPV)

SmartAPI supports ordering of Mortgage Participant Verification (MPV) reports. This product is meant to be used to check loan participants (e.g. loan officer, brokerage, appraiser, etc.) against various watchlists. It is not intended to substitute a comprehensive fraud product that would typically be ordered for borrowers of a loan.

WORKFLOW

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PACKAGED REPORT

The Packaged Report is a fraud detection product that supports verification across three types of entities typically involved in a loan: the borrower(s), the subject property, and any non-borrower participants (e.g. loan officer, underwriter, etc). Various services are used to generate these reports, some of which include: Lexis Nexis (WsAccurint, WsIdentity, Bridger Insight XG), Equifax Intellireal, Experian Income Insights/Income Insights W2, MERS, MCL Integration (UDN), CreditData and watchlists.

Packaged Reports are organized into templates, which we commonly refer to as layouts. Each layout will have its own customization of products to be ordered for each of the three entities. When requesting a new packaged report, you need only indicate the layout to be ordered and include information for the entities that will be involved. Whether a borrower, the subject property, or a participant needs to be included in your request will depend on the layout being ordered. Therefore, it is important to communicate with the service provider or your end-user regarding what layouts are planned for use. Alternatively, you can include information for all three entities for all packaged report orders, thus ensuring there will always be enough information to process the request, regardless of layout.

When you begin testing the packaged reports, please notify your POC to enable the ALL_PRODUCTS layout and any additional specified layouts if needed

WORKFLOW

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- **Processing**: The system is continuing to process your order. Continue to poll for the VendorOrderIdentifier until you receive an end status.
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POLLING GUIDELINES

The following polling rules should be adhered to:

- If the VendorOrderIdentifier is not in an end status after 270 seconds from the time the new order request was submitted, you should time out the order and instruct the end-user to reach out to their service provider. These orders should not take this long to process, so the system is likely experiencing an issue.

GET PACKAGED REPORT LAYOUTS

Each client will typically have a different set of packaged report layouts available to them. Since the report layout name is something that must be included with the order, instead of storing and maintaining a table of these within your system, you could query our system for this list each time the user goes to order a packaged report. These layout names could then be populated into a list from which the user can select the layout they want ordered. An XML sample is included for this method.

PACKAGED REPORT LAYOUT EXAMPLES

Below are some examples of packaged report layouts:

The below layout contains one product for each entity or section. This means at least one borrower, one subject property, and one participant will need to be included with the order.

Section	Product
Borrower	Instant ID
Property	House Price Index
Participant	Watchlist for ASC National Registry of Appraisers

The below layout contains two services for the borrower section. The other 2 sections have been intentionally left blank. This means at least one borrower will need to be included with the request. If a subject property and participant are included, they will simply be ignored, since there is no product configured for them.

Section	Product
Borrower	Person Search, Liens and Judgements
Property	
Participant	

UPGRADING A PACKAGED REPORT

After a packaged report is completed, the user has the option of upgrading it. In this scenario, you have a packaged report where you either need to add a coborrower, a subject properties and/or additional participants. You can even upgrade to a larger layout that has more products configured (larger layout must fully overlap all products on the existing file). The upgrade request must contain the VendorOrderIdentifier of the existing packaged report you intend to upgrade along with the new entities (borrower/participants) or asset/collateral (subject property) you will be adding to it. Additionally, you can submit an upgrade request without adding any new elements to the request which will essentially refresh the existing packaged report. Please note that this will not reorder any completed/existing products but will attempt to reorder products if they failed or errored on the original packaged report.

When submitting an upgrade request, the file will move into a processing status, at which point you'll want to poll as you would on a new order.

PRECISE ID KNOWLEDGE IQ

SmartAPI supports ordering of knowledge-based authentication products. This product will return a series of questions that the applicant must answer. Once the answers have been submitted, a completed report will be returned. SmartAPI Precise ID orders will also support passing customer level credentials in the request itself. Please reference the advanced sample XML on where to place the ServiceLogin and ServicePassword fields.

WORKFLOW

When a new order is submitted (or a query of an existing order), the response XML will include a StatusCode value that appears under the //SERVICE/STATUSES/STATUS container. That StatusCode will dictate your required follow-up action. Below are the appropriate actions to take, based on the StatusCode returned:

- **New:** The system received your recent order request and is processing it. A unique VendorOrderIdentifier was generated for this order and you should begin polling for that until you receive an end status.
- **Pending**: The system is awaiting your answers or is processing your answers.
    - On a new order request, getting back this status means your order was processed successfully and the questions and VendorOrderIdentifier will be included in the response. You should present the questions to the user. Once the user submits their answer, you'll want to reference the VendorOrderIdentifier in your "answer" request.
    - On an "answer" request, this status means your answers were successfully accepted by our system and we are awaiting a response from the vendor to get a completed report. You should begin polling until you get an end status.
- **Completed**: The order is complete and the data you've requested is included in the response. This is considered an end status.
- **Error**: An error occurred while processing the request. A description should be included under the StatusDescription element. Please display this to the end-user. This is considered an end status.

POLLING GUIDELINES

The following polling rules should be adhered to:

- If you do not get an end status within 90 seconds of submitting your answers, you should time out your order and instruct the user to contact the service provider. Once answers are submitted, getting a completed report should not take more than a few seconds, so a delay of this length indicates a system issue.
- If you do not get a Pending status within 90 seconds of a new order, you should time out the order and instruct the user to contact the service provider. There is likely an issue with our system or our connection to the vendor.
- Users will have approximately ten minutes to submit answers. The actual time will vary by a few minutes for each user, but "approximately 10 minutes" is a safe phrase to display to your user when notifying them of a time limit. If you attempt to submit answers after the user's specific time window has passed, we will return an Error. If you poll for an order that has timed out, we will return a Completed status, but no report will be included with the response.
- Answers should be submitted in the same sequence in which they appear in the SmartAPI response.

PRECISE ID SCORE

SmartAPI supports ordering of Experian Precise ID Score. Precise ID Score will return a report with a score. SmartAPI Precise ID orders will also support passing customer level credentials in the request itself. Please reference the advanced sample on where to place the ServiceLogin and ServicePassword fields.

WORKFLOW

When a new order is submitted (or a query of an existing order), the response XML will include a StatusCode value that appears under the //SERVICE/STATUSES/STATUS container. That StatusCode will dictate your required follow-up action. Below are the appropriate actions to take, based on the StatusCode returned:

- **New**: The system received your recent order request and is processing it. A unique VendorOrderIdentifier was generated for this order and you should begin polling for that until you receive an end status.
- **Processing**: The system is continuing to process your order. Continue to poll for the VendorOrderIdentifier until you receive an end status.
- **Completed**: The order is complete and the data you've requested is included in the response. This is considered an end status.
- **Error**: An error occurred while processing the order. A description should be included under the StatusDescription element. Please display this to the end-user. This is considered an end status.

POLLING GUIDELINES

The following polling rules should be adhered to:

- If the VendorOrderIdentifier is not in an end status after 90 seconds from the time the new order request was submitted, you should time out the order and instruct the end-user to reach out to their service provider. These orders should not take this long to process, so the system is likely experiencing an issue.

SSA-89

SmartAPI supports order of SSA-89s.

WORKFLOW

When a new order is submitted (or a query of an existing order), the response XML will include a StatusCode value that appears under the //SERVICE/STATUSES/STATUS container. That StatusCode will dictate your required follow-up action. Below are the appropriate actions to take, based on the StatusCode returned:

- **New**: The system received your recent order request and is processing it. A unique VendorOrderIdentifier was generated for this order and you should begin polling for that until you receive an end status.
- **Pending**: The system received your recent order request and is processing it. A unique VendorOrderIdentifier was generated for this order and you should begin polling for that until you receive an end status.
- **Processing**: The system is continuing to process your order. Continue to poll for the VendorOrderIdentifier until you receive an end status.
- **Completed**: The order is complete and the data you've requested is included in the response. This is considered an end status.
- **Error**: An error occurred while processing the order. A description should be included under the StatusDescription element. Please display this to the end-user. In some cases, a document might also be included with the response if you requested one. Please check for the presence of a document and make it available to the user for viewing if present. This is considered an end status.
- **Canceled**: The order was canceled by the vendor. This is typically done at the request of the end-user. This is considered an end status.
- **Rejected**: The order was rejected by the vendor. Please check the StatusDescription for an explanation and display it to the end-user. Common reasons are: SSA89 authorization form not properly filled out; outdated version of SSA89 authorization form was submitted. This is considered an end status.

POLLING GUIDELINES

The following polling rules should be adhered to:

- If the VendorOrderIdentifier is not in an end status after 30 seconds from the time the new order request was submitted, your order is being manually reviewed and may take several days to complete. We recommend automatically polling for it once or twice a day to ensure the status of the order on your end is in sync with our system.
- We recommend allowing the user to manually submit a retrieval request on orders that are being manually reviewed. This will allow them to immediately retrieve the completed report once they are notified of its completion, rather than waiting for the next polling run.

TAX TRANSCRIPT (TAX)

Commonly referred to as 4506-C, SmartAPI supports ordering of tax transcripts and other tax-related forms.

WORKFLOW

When a new order is submitted (or a query of an existing order), the response XML will include a StatusCode value that appears under the //SERVICE/STATUSES/STATUS container. That StatusCode will dictate your required follow-up action. Below are the appropriate actions to take, based on the StatusCode returned:

- **New**: The system received your recent order request and is processing it. A unique VendorOrderIdentifier was generated for this order and you should begin polling for that until you receive an end status.
- **Pending**: The system received your recent order request and is processing it. A unique VendorOrderIdentifier was generated for this order and you should begin polling for that until you receive an end status.
- **Processing**: The system is continuing to process your order. Continue to poll for the VendorOrderIdentifier until you receive an end status.
- **Completed**: The order is complete and the data you've requested is included in the response. This is considered an end status.
- **Error**: An error occurred while processing the order. A description should be included under the StatusDescription element. Please display this to the end-user. This is considered an end status.
- **Canceled**: The order was canceled by the vendor/IRS. Please check the StatusDescription for an explanation and display it to the end-user. This is considered an end status.
- **Rejected**: The order was rejected by the vendor/IRS. Please check the StatusDescription for an explanation and display it to the end-user. Canceled and Rejected are similar statuses with the difference being Rejected usually results in a charge being billed to the user. This is considered an end status.

POLLING GUIDELINES

The following polling rules should be adhered to:

- If the VendorOrderIdentifier is not in an end status after 30 seconds from the time the new order request was submitted, your order is being processed manually and it will take several days to complete. We recommend automatically polling for it once or twice a day to ensure the status of the order on your end is in sync with our system.
- We recommend allowing the user to manually submit a retrieval request on orders that are being manually processed. This will allow them to immediately retrieve the completed report once they are notified of its completion, rather than waiting for the next polling run.

PRODUCT FEATURES

SUPPORTED FORMS

The IRS has many tax documents that can be filed. Our system supports the most common ones, listed below:

- W2: Used by businesses to report an employee's income and other income-related details.
- 1099: Used by businesses to report a contractor's income and income-related details.
- 1040: Used by individuals to file annual income tax returns.
- 1065: Used by partnerships to file annual income tax returns.
- 1120: Used by corporations to file annual income tax returns.

The table below shows the enumerations used to reference each of these documents and their corresponding transcript sub-options. Exactly one tax form and corresponding sub-option is to be selected for each request. This is done by setting the tax form type under the IRSDocumentType element and setting "true" on the transcript sub-option you want to order. If more than one sub-option is set to "true", SmartAPI will use the first option it parses and ignore the rest. Please review the XML samples for examples.

Tax Form	Supported Transcript Types
IRSW2	OtherTranscriptDescription
IRS1099	OtherTranscriptDescription
IRS1040	ReturnTranscriptIndicator, AccountTranscriptIndicator, RecordOfAccountIndicator, VerificationOfNonfilingIndicator
IRS1065	ReturnTranscriptIndicator, AccountTranscriptIndicator, RecordOfAccountIndicator, VerificationOfNonfilingIndicator
IRS1120	ReturnTranscriptIndicator, AccountTranscriptIndicator, RecordOfAccountIndicator, VerificationOfNonfilingIndicator

FANNIE MAE DAY 1 CERTAINTY

Fannie Mae's Day 1 Certainty (D1C) program allows customers to submit tax return related information to Fannie Mae for underwriting. Customers or underwriters will need a transcript provider D1C reference number that they can enter in Fannie Mae to import a transcript.

If your transcript provider has an integration with Fannie Mae D1C, they can pass their D1C reference number to MCL, which will then be available to you via SmartAPI. This functionality allows you to automate submitting transcripts to Fannie Mae for underwriting purposes. The information will be provided in the response to a retrieve request. The D1C related information will be available in the following elements:

Data Field	MISMO XPATH	Description
Product Identifier	//SERVICE_PRODUCT_IDENTIFIERS/SERVICE_PRODUCT_IDENTIFIER/SERVICE_PRODUCT_IDENTIFIER_DETAIL/ProductIdentifier	Identifies the transcript provider's D1C reference number.
Product Identifier Type	//SERVICE_PRODUCT_IDENTIFIERS/SERVICE_PRODUCT_IDENTIFIER/SERVICE_PRODUCT_IDENTIFIER_DETAIL/ProductIdentifierType="AUSReferenceNumber"	Identifies what type of ID the *ProductIdentifier* field is. For D1C, this value will be "AUSReferenceNumber".
Source Name	//SERVICE_PRODUCT_IDENTIFIERS/SERVICE_PRODUCT_IDENTIFIER/SERVICE_PRODUCT_IDENTIFIER_DETAIL/SourceName	Identifies who is providing the *ProductIdentifier* value.
Source Identifier	//SERVICE_PRODUCT_IDENTIFIERS/SERVICE_PRODUCT_IDENTIFIER/SERVICE_PRODUCT_IDENTIFIER_DETAIL/SourceIdentifier	Unique numerical identifier tied to the *SourceName*. This field makes it easier to parse and match sources if using multiple sources.

UNDISCLOSED DEBT NOTIFICATION (UDN)

SmartAPI supports ordering of Undisclosed Debt Notifications. This product is a composition of the Equifax Undisclosed Debt Monitoring, Experian Mortgage Watch, and Transunion Pre-Closing Mortgage Notifications products.

HIGH-LEVEL OVERVIEW

The UDN file that is generated for each order request is a container that holds the bureaus' monitoring products and so does not have its own status in the same way we'd see with other products offered through SmartAPI. However, each bureau is processed independently and will have its own status within the UDN file.

Since each bureau within the UDN container is processed independently, it's possible for the status of each bureau to differ between one another. For example, some bureaus may successfully activate, while others may error. The status for each bureau will be sent in the response.

Our system supports joint UDN orders, which are instances where a single UDN order ID contains two persons. To create these, submit the initial activation order as a joint (see sample xmls). We recommend this approach, as it allows users to take advantage of preferred pricing on joint orders. However, for polling and deactivation requests, those will need to be done one person at a time. That is, on joint UDN files, polling or deactivation requests for each person is done as a separate request.

Once monitoring is active, any alerts received from the bureaus will be sent to the notification e-mail addresses that were included with the UDN order. The end-user could then retrieve the UDN report to view the full details of every alert. The UDN report is cumulative, so it will contain all alerts that have been received on the file.

The user does not have the ability to specify which bureaus they'd like to include with their UDN order. When a new order request is submitted, whatever the service provider has configured for that user is what gets ordered. Therefore, you will see that there is no way to specify which bureaus are to be included on an order request.

WORKFLOW

When a request is submitted, the StatusCode value that appears under the //SERVICE/STATUSES/STATUS container will dictate whether your submission was successfully processed or failed. Below are the appropriate actions to take, based on the nature of your request and the StatusCode returned.

NEW ORDER OR RETRIEVE EXISTING ORDER

- **Error**:
  - If there exists at least one //SERVICE/CREDIT/CREDIT_REQUEST/CREDIT_INQUIRIES/CREDIT_INQUIRY/CREDIT_REPOSITORIES/CREDIT_REPOSITORY node in the response, then process as you would if the StatusCode was a null, empty string, or anything other than "Error".
  - If there does not exist at least one //SERVICE/CREDIT/CREDIT_REQUEST/CREDIT_INQUIRIES/CREDIT_INQUIRY/CREDIT_REPOSITORIES/CREDIT_REPOSITORY node, then the system was unable to process your request. A reason should appear under the StatusDescription sibling element. Please parse this and display to the user.
- **Null, empty string or anything other than "Error"**: The system successfully processed your request. If your request was for a new order, a VendorOrderIdentifier would be included in the response. If your request was for a retrieval of an existing file, then the data you requested would be included in the response.

UPDATE EXISTING ORDER

This method allows you to update the notification emails on file for an existing order. Note the new emails provided will replace what is currently stored. If you need to add an additional email to what is stored, please include what is currently stored along with the new email(s) in your request.

- **Error**: Your request was not processed successfully. Please see the corresponding StatusDescription for the reason.
- **Null, empty string or anything other than "Error"**: The system successfully processed your request.

DEACTIVATE EXISTING ORDER

This method is used to initiate deactivation of the order (i.e. request that bureaus stop monitoring the consumer). Note that a deactivation request will only be accepted if at least 1 bureau is in Active status. This means you will receive an error if the deactivation request is sent on an order that is still attempting to activate, is in the process of deactivating, or similar.

- **Error**: Your request was not processed successfully. Please see the corresponding StatusDescription for the reason.
- **Null, empty string or anything other than "Error"**: The system successfully processed your request.

POLLING GUIDELINES

The following polling rules should be adhered to:

- We recommend automatically polling all active orders once or twice a day to ensure the status of each bureau from your end is in sync with what our system has. This should continue until all bureaus are in an end status.
- We recommend allowing the user to manually submit a poll/retrieval request for the order. This would typically be done when they receive an email notification of a new alert. Allowing them to retrieve the details of the report on-demand will allow them to view the details of the alert without waiting for your next polling run.
- On joint UDN files, polling or deactivation requests for each person is done as a separate request.
- When retrieving an existing file, usable data (i.e. parsable XML or an HTML/PDF file) will only be available if at least one bureau was successfully activated. This does not mean at least one bureau must be in Active status, it just means one bureau must have been Active at one point.
- When submitting a deactivation request, you should follow up with a "retrieve existing" request to grab the updated statuses for each bureau on the UDN file. If your deactivation request was processed successfully, at least one bureau should now be showing Deactivating status.

BUREAU STATUSES

The below enumerates the statuses that each bureau of the UDN order can exhibit. Each bureau's status can be found under the following node:
//SERVICE/CREDIT/CREDIT_REQUEST/CREDIT_INQUIRIES/CREDIT_INQUIRY/CREDIT_REPOSITORIES//CREDIT_REPOSITORY/EXTENSION/OTHER/p3:Status

- **Processing**: Your activation request is being processed. Turnaround time is typically 24-48 hours.
- **Active**: The borrower's credit is being monitored by the bureau. New alerts will appear on the file and corresponding emails sent to any Notification Emails associated with the order.
- **Deactivating**: Your deactivation request is being processed. Turnaround time is typically 24-48 hours.
- **Deactivated**: The borrower's credit is no longer being monitored. This is considered an end status.
- **Error**: An error occurred with the bureau. A sibling node of p3:ErrorMessage may be present with more details of the error. This is considered an end status.

RESPONSE TABLE

Data Field	MISMO XPATH	Description

Bureau Status	//SERVICES/SERVICE/CREDIT/CREDIT_REQUEST/CREDIT_INQUIRIES/CREDIT_INQUIRY/CREDIT_REPOSITORIES/CREDIT_REPOSITORY/EXTENSION/OTHER/Status	Describes current bureau status, which may be Processing, Error, Active, Deactivating, and Deactivated. There will be one status per bureau that is ordered.
Bureau Error Messages	//SERVICES/SERVICE/CREDIT/CREDIT_REQUEST/CREDIT_INQUIRIES/CREDIT_INQUIRY/CREDIT_REPOSITORIES/CREDIT_REPOSITORY/EXTENSION/OTHER/ErrorMessage	Displays bureau error message.
UDN Start Date	//SERVICES/SERVICE/CREDIT/CREDIT_REQUEST/CREDIT_INQUIRIES/CREDIT_INQUIRY/CREDIT_REPOSITORIES/CREDIT_REPOSITORY/EXTENSION/OTHER/StartDate	Date UDN order is to begin monitoring credit. There will be one Start Date per bureau that is ordered.
UDN End Date	//SERVICES/SERVICE/CREDIT/CREDIT_REQUEST/CREDIT_INQUIRIES/CREDIT_INQUIRY/CREDIT_REPOSITORIES/CREDIT_REPOSITORY/EXTENSION/OTHER/EndDate	Date UDN order is set to stop monitoring credit. There will be one End Date per bureau that is ordered.
[TRADELINE INFORMATION]		
Source Bureau	//SERVICE/CREDIT/CREDIT_RESPONSE/CREDIT_LIABILITIES/CREDIT_LIABILITY/CREDIT_REPOSITORIES/CREDIT_REPOSITORY/CreditRepositorySourceType	
Notification Type	//SERVICE/CREDIT/CREDIT_RESPONSE/CREDIT_LIABILITIES/CREDIT_LIABILITY/EXTENSION/OTHER/NotificationType	Tradelines, Delinquencies, Major Derogatory tradelines, Tradeline Bankruptcies, Scheduled Payment
Notification Date	//SERVICE/CREDIT/CREDIT_RESPONSE/CREDIT_LIABILITIES/CREDIT_LIABILITY/EXTENSION/OTHER/NotificationDate	
Status	//SERVICE/CREDIT/CREDIT_RESPONSE/CREDIT_LIABILITIES/CREDIT_LIABILITY/CREDIT_LIABILITY_CURRENT_RATING/CreditLiabilityCurrentRatingType	Tradeline status rating; Current, Derogatory, etc.
Creditor Name	//SERVICE/CREDIT/CREDIT_RESPONSE/CREDIT_LIABILITIES/CREDIT_LIABILITY/CREDIT_LIABILITY_CREDITOR/NAME/FullName	
Account Type	//SERVICE/CREDIT/CREDIT_RESPONSE/CREDIT_LIABILITIES/CREDIT_LIABILITY/CREDIT_LIABILITY_DETAIL/CreditLiabilityAccountType	
Opened Date	//SERVICE/CREDIT/CREDIT_RESPONSE/CREDIT_LIABILITIES/CREDIT_LIABILITY/CREDIT_LIABILITY_DETAIL/CreditLiabilityAccountOpenedDate	
Reported Date	//SERVICE/CREDIT/CREDIT_RESPONSE/CREDIT_LIABILITIES/CREDIT_LIABILITY/CREDIT_LIABILITY_DETAIL/CreditLiabilityAccountReportedDate	
Balance	//SERVICE/CREDIT/CREDIT_RESPONSE/CREDIT_LIABILITIES/CREDIT_LIABILITY/CREDIT_LIABILITY_DETAIL/CreditLiabilityUnpaidBalanceAmount	
Terms	//SERVICE/CREDIT/CREDIT_RESPONSE/CREDIT_LIABILITIES/CREDIT_LIABILITY/CREDIT_LIABILITY_DETAIL/CreditLiabilityTermsDescription	
Credit Limit	//SERVICE/CREDIT/CREDIT_RESPONSE/CREDIT_LIABILITIES/CREDIT_LIABILITY/CREDIT_LIABILITY_DETAIL/CreditLiabilityCreditLimitAmount	
Industry	//SERVICE/CREDIT/CREDIT_RESPONSE/CREDIT_LIABILITIES/CREDIT_LIABILITY/CREDIT_LIABILITY_DETAIL/CreditBusinessType	
Remark	/MESSAGE/DEAL_SETS/DEAL_SET/DEALS/DEAL/SERVICES/SERVICE/CREDIT/CREDIT_RESPONSE/CREDIT_LIABILITIES/CREDIT_LIABILITY/CREDIT_COMMENTS/CREDIT_COMMENT	This container has the credit comment source type, comment, and comment type.
Bankruptcy Status	//SERVICE/CREDIT/CREDIT_RESPONSE/CREDIT_LIABILITIES/CREDIT_LIABILITY/EXTENSION/OTHER/TRADELINE_BANKRUPTCY/BankruptcyStatus	Only applicable to bankruptcy tradelines.
Bankruptcy Type	//SERVICE/CREDIT/CREDIT_RESPONSE/CREDIT_LIABILITIES/CREDIT_LIABILITY/EXTENSION/OTHER/TRADELINE_BANKRUPTCY/BankruptcyType	Only applicable to bankruptcy tradelines.
[INQUIRIES INFORMATION]		
Source Bureau	//SERVICE/CREDIT/CREDIT_RESPONSE/CREDIT_LIABILITIES/CREDIT_LIABILITY/CREDIT_REPOSITORIES/CREDIT_REPOSITORY/CreditRepositorySourceType	

Notification Type	N/A	This is derived from the parent container, which is CREDIT_INQUIRY.
Notification Date	//SERVICE/CREDIT/CREDIT_RESPONSE/CREDIT_LIABILITIES/CREDIT_LIABILITY/EXTENSION/OTHER/NotificationDate	
Company	//SERVICE/CREDIT/CREDIT_RESPONSE/CREDIT_INQUIRIES/CREDIT_INQUIRY/NAME/FullName	
Inquiry Date	//SERVICE/CREDIT/CREDIT_RESPONSE/CREDIT_INQUIRIES/CREDIT_INQUIRY/CREDIT_INQUIRY_DETAIL/CreditInquiryDate	
Industry	//SERVICE/CREDIT/CREDIT_RESPONSE/CREDIT_INQUIRIES/CREDIT_INQUIRY/CREDIT_INQUIRY_DETAIL/CreditBusinessType	
Permissible Purpose	//SERVICE/CREDIT/CREDIT_RESPONSE/CREDIT_INQUIRIES/CREDIT_INQUIRY/EXTENSION/OTHER/PermissiblePurpose	
Reissue Date	//SERVICE/CREDIT/CREDIT_RESPONSE/CREDIT_INQUIRIES/CREDIT_INQUIRY/EXTENSION/OTHER/ReissueDate	
Remark	//SERVICE/CREDIT/CREDIT_RESPONSE/CREDIT_INQUIRIES/CREDIT_INQUIRY/CREDIT_COMMENTS/CREDIT_COMMENT/CreditCommentText	
[PUBLIC RECORDS INFORMATION]		
Source Bureau	//SERVICE/CREDIT/CREDIT_RESPONSE/CREDIT_LIABILITIES/CREDIT_LIABILITY/CREDIT_REPOSITORIES/CREDIT_REPOSITORY/CreditRepositorySourceType	
Notification Date	//SERVICE/CREDIT/CREDIT_RESPONSE/CREDIT_LIABILITIES/CREDIT_LIABILITY/EXTENSION/OTHER/NotificationDate	
Court Info	//SERVICE/CREDIT/CREDIT_RESPONSE/CREDIT_PUBLIC_RECORDS/CREDIT_PUBLIC_RECORD/CREDIT_PUBLIC_RECORD_DETAIL/CreditPublicRecordCourtName	
Record Type	//SERVICE/CREDIT/CREDIT_RESPONSE/CREDIT_PUBLIC_RECORDS/CREDIT_PUBLIC_RECORD/CREDIT_PUBLIC_RECORD_DETAIL/CreditPublicRecordType	
ECOA	//SERVICE/CREDIT/CREDIT_RESPONSE/CREDIT_PUBLIC_RECORDS/CREDIT_PUBLIC_RECORD/CREDIT_PUBLIC_RECORD_DETAIL/CreditPublicRecordAccountOwnershipType	
File Date	//SERVICE/CREDIT/CREDIT_RESPONSE/CREDIT_PUBLIC_RECORDS/CREDIT_PUBLIC_RECORD/CREDIT_PUBLIC_RECORD_DETAIL/CreditPublicRecordFiledDate	
Release Date	//SERVICE/CREDIT/CREDIT_RESPONSE/CREDIT_PUBLIC_RECORDS/CREDIT_PUBLIC_RECORD/EXTENSION/OTHER/ReleaseDate	
Liability Amount	//SERVICE/CREDIT/CREDIT_RESPONSE/CREDIT_PUBLIC_RECORDS/CREDIT_PUBLIC_RECORD/CREDIT_PUBLIC_RECORD_DETAIL/CreditPublicRecordLegalObligationAmount	
Asset Amount	//SERVICE/CREDIT/CREDIT_RESPONSE/CREDIT_PUBLIC_RECORDS/CREDIT_PUBLIC_RECORD/EXTENSION/OTHER/AssetAmount	
Industry	//SERVICE/CREDIT/CREDIT_RESPONSE/CREDIT_PUBLIC_RECORDS/CREDIT_PUBLIC_RECORD/CREDIT_PUBLIC_RECORD_DETAIL/Industry	
[HIGH UTILIZATION INFORMATION]		
Source Bureau	//SERVICE/CREDIT/CREDIT_RESPONSE/CREDIT_LIABILITIES/CREDIT_LIABILITY/CREDIT_REPOSITORIES/CREDIT_REPOSITORY/CreditRepositorySourceType	
Notification Date	//SERVICE/CREDIT/CREDIT_RESPONSE/CREDIT_LIABILITIES/CREDIT_LIABILITY/EXTENSION/OTHER/NotificationDate	
Creditor Name	//SERVICE/CREDIT/CREDIT_RESPONSE/CREDIT_LIABILITIES/CREDIT_LIABILITY/CREDIT_LIABILITY_CREDITOR/NAME/FullName	
Balance	//SERVICE/CREDIT/CREDIT_RESPONSE/CREDIT_LIABILITIES/CREDIT_LIABILITY/CREDIT_LIABILITY_DETAIL/CreditLiabilityUnpaidBalanceAmount	
Industry	//SERVICE/CREDIT/CREDIT_RESPONSE/CREDIT_LIABILITIES/CREDIT_LIABILITY/CREDIT_LIABILITY_DETAIL/CreditBusinessType	

Type	//SERVICE/CREDIT/CREDIT_RESPONSE/CREDIT_LIABILITIES/CREDIT_LIABILITY/EXTENSION/OTHER/HighUtilizationType
Reported Date	//SERVICE/CREDIT/CREDIT_RESPONSE/CREDIT_LIABILITIES/CREDIT_LIABILITY/CREDIT_LIABILITY_DETAIL/CreditLiabilityAccountReportedDate

VERIFICATION OF ASSET (VOA)

SmartAPI offers asset verifications through borrower-managed account enrollment. Upon successfully placing an order, an e-mail will be sent to the borrower. The borrower must follow the instructions in the e-mail to authorize access to their accounts. Once enrollment is complete, the user will be able to pull down a report detailing balance and transaction history for all enrolled accounts.

HIGH-LEVEL OVERVIEW

Below is a high-level workflow overview of the VOA product.

1. **New order is placed**: New order is submitted to SmartAPI. SmartAPI validates the order. If successful, it is passed on to the 3rd party vendor. The vendor sends the borrower a "welcome" email if the order was successful.
2. **Borrower enrolls financial accounts**: The "welcome" email contains a link the borrower can click to be taken to the vendor's website, where they will be able to enroll their various asset accounts to be shared. Once enrollment is complete, the user will receive an email letting them know that the borrower has completed the enrollment process.
3. **Order goes into Active status:** When enrollment is complete, the order will go into Active status.
4. **User refreshes and retrieves updated asset report and repeats, as needed**: When the user wishes to retrieve an updated asset report, they will submit a "refresh" request. SmartAPI will then reach out to the vendor to download the latest asset report for the order and pull that into our system. If the refresh request is successful, the new asset report will be available for retrieving shortly afterwards. Your system can send refresh requests for as long as the refresh period is open.

WORKFLOW

When a request is submitted, the StatusCode value that appears under the //SERVICE/STATUSES/STATUS container will dictate whether your submission was successfully processed or if it failed. Below are the appropriate actions to take, based on the nature of your request and the StatusCode returned.

NEW ORDER OR RETRIEVE EXISTING ORDER

- **New**: The system received your recent order request and is processing it. A unique VendorOrderIdentifier was generated for this order and should appear in the response. A StatusDescription sibling node may be present; if so, it will contain a description of the order's current status. You may begin polling for this order.
- **Pending**: The borrower is in the process of enrolling their financial accounts. A sibling node of StatusDescription should be present with a description of the order's current status.
- **Active**: The borrower successfully enrolled at least one account. Refreshing of the asset report is an available action. The latest asset report on file will also be included with the response, if one exists.
- **Alert**: The order is active, but has encountered an issue. One reason this can happen is if the credentials the borrower initially provided to the vendor no longer match what their financial institution has on file. A StatusDescription sibling node should be present, which contains more details of the issue. The latest asset report on file will also be included with the response, if one exists. Refreshing of the asset report is an available action.
- **Completed**: The order is closed and access to the borrower's financial accounts has been removed. The latest asset report on file will also be included with the response, if one exists. This is considered an end status.
- **Error**: The order encountered an error. A StatusDescription sibling node should be available with more details. This is considered an end status.
- **Canceled**: The order is closed and borrower never enrolled any accounts. This is considered an end status.

REFRESH EXISTING ORDER

- **Error**: Your request was not processed successfully. Please see the corresponding StatusDescription for the reason.

- **Anything other than "Error"**: The system will normally echo back the status of the order—as opposed to the status of your refresh request—if no error was encountered with your refresh request. Therefore, if you receive a status of anything other than "Error", it means the system processed your refresh request successfully.

POLLING GUIDELINES

The following polling rules should be adhered to:

- From the moment a new order is submitted to the time it goes into an end status, we recommend polling for it once or twice a day to keep the status of the order on your end in sync with what we have.
- You should allow the user to manually submit a refresh request on orders that are in or have at one point gone into Active status. This is especially important when the borrower completes enrollment and the user is sent an email notification. The user would normally want to generate the asset report right away, so that they can review the borrower's financial data.
- When a refresh request is initiated, your system should automatically follow up with polling requests in short intervals. The p3:ReportStatus returned in the polling response will initially show "InProgress", while our system downloads the latest asset report from the vendor. Once the p3:ReportStatus shows "Ready", the latest asset report we retrieved will also be included in the response. This entire process should be relatively quick. If 300 seconds has passed and the p3:ReportStatus is still in "InProgress" status, please display a timeout message to the user and instruct them to contact the service provider.
- Continuing off the previous point, we recommend allowing the user to manually submit a retrieve request to obtain the latest asset report on file. In the event that a refresh request is taking longer than usual to process, once it does complete, the user needs to be given a way to retrieve it.

ASSET REPORT STATUS

The following node, present when retrieving existing orders, is used to determine if the asset report is currently being refreshed:
//SERVICE/SERVICE_PRODUCT_FULFILLMENT/SERVICE_PRODUCT_FULFILLMENT_DETAIL/EXTENSION/OTHER/p3:ReportStatus

- **InProgress**: The system is currently refreshing the asset report.
- **Ready**: The asset report is not currently being refreshed. If you recently requested a refresh, this means the latest asset report was downloaded from the vendor and is included with the response.

REFRESH PERIODS

SmartAPI supports the following enumerations for refresh periods. Not all service providers will accept all enumerations. Flexibility should be given to your user in selecting the refresh period to ensure they can pass a value that will be accepted by their service provider.

- **None**: The asset report can be refreshed for only a short duration, usually a few days. The length will vary between providers; the idea is to allow the borrower a few days to enroll all their accounts, as well as allow the user to retrieve the updated asset report as accounts are enrolled, before shutting off access to the accounts.
- **_30Days**: The asset report can be refreshed for the next 30 days.
- **_60Days**: The asset report can be refreshed for the next 60 days.
- **_90Days**: The asset report can be refreshed for the next 90 days.

There is no limit on the number of refreshes that can be done while an order is active. However, data is typically pulled from financial institutions only once per day.

VERIFICATION OF DEPOSIT (VOD)

SmartAPI supports deposit account verifications, which are manually processed by the verification service provider.

WORKFLOW

When a new order is submitted (or a query of an existing order), the response XML will include a StatusCode value that appears under the //SERVICE/STATUSES/STATUS container. That StatusCode will dictate your required follow-up action. Below are the appropriate actions to take, based on the StatusCode returned:

- **New**: The system received your recent order request and is processing it. A unique VendorOrderIdentifier was generated for this order and you should use that when polling.
- **Pending**: Your order is still being processed.
- **Completed**: The order is complete and the data you've requested is included in the response. This is considered an end status.
- **Canceled**: Your order was canceled by the service provider. This is considered an end status.
- **Error**: Your request could not be completed. A StatusDescription sibling node will accompany this with more details. This is considered an end status.

POLLING GUIDELINES

The following polling rules should be adhered to:

- From the time a new order is successfully submitted until it reaches an end status, we recommend polling for it once or twice a day to ensure the status of the order on your end is in sync with our system. VOD orders typically take several days to complete.
- We recommend allowing the user to manually submit a retrieval request. This will allow them to immediately retrieve the completed VOD report once they are notified of its completion, rather than waiting for the next polling run.

VERIFICATION OF EMPLOYMENT (VOE)

SmartAPI supports verification of employment (VOE) services through instant database lookups and manual verifications. VOE orders can be processed instantly if the applicant's employer reports information to automated services like The Work Number and Experian Verify. For all other employers, the VOE order will be processed manually.

WORKFLOW

When a new order is submitted (or a query of an existing order), the response XML will include a StatusCode value that appears under the //SERVICE/STATUSES/STATUS container. That StatusCode will dictate your required follow-up action. Below are the appropriate actions to take, based on the StatusCode returned:

- **New**: The system received your recent order request and is processing it. A unique VendorOrderIdentifier was generated for this order and you should begin polling for that until you receive an end status.
- **Processing**: The system is continuing to process your order. Continue to poll for the VendorOrderIdentifier until you receive an end status.
- **Completed**: The order is complete and the data you've requested is included in the response. This is considered an end status.
- **Canceled**: Your order was canceled. Order cancellations are typically done at the direction of the end-user. You can convey this status to the end-user and advise them to contact the service provider with any questions. This is considered an end status.
- **Error**: An error occurred while processing the order. A description should be included under the StatusDescription element. Please display this to the end-user. This is considered an end status.
- **NoHit**: This status is often seen on instant VOE orders. This means the applicant was not located via instant database lookup. The end-user should resubmit as a manual order if they are confident that they have the borrower's information correct. This is considered an end status.

POLLING INSTRUCTIONS

The following polling rules should be adhered to:

- If the VendorOrderIdentifier is not in an end status after 30 seconds from the time the new order request was submitted, your order is likely being processed manually and it will take several days to weeks for completion, depending on the employer's responsiveness. We recommend automatically polling for it once or twice a day to ensure the status of the order on your end is in sync with our system.
- We recommend allowing the user to manually submit a retrieval request on orders that are being manually processed. This will allow them to immediately retrieve the completed VOE report once they are notified of its completion, rather than waiting for the next polling run.

VERIFICATION OF EMPLOYMENT PRODUCTS

The below table lists the types of verifications that are supported. The column definitions are as follows:

- **ID**: These ID values correspond to the XML samples provided.
- **Product:** The name with which the product can generally be referenced. Some of these are generic, some of these are branded by the vendor.
- **Fulfillment:** Describes the method used to fulfil the order.
- **Employer Filter:** The type of employment data that the product will return (e.g. current employers, previous employers, etc).
- **Employer Info:** Describes whether specific employer information can be included with the order.

- **Description:** A description of what the product is.
- **Vendor:** Describes which vendor(s) each product option is fulfilled by. This is for reference only; there is not a place in the XML request to specify the vendor to fulfill the order. Instead, the vendor will be determined by the configuration of the account settings.

You may notice that many of these products are very similar. This is due to minor differences in requirements among vendors. Integrators who are hesitant to support all these products should consult with their user base to determine what verification products they will need, at minimum. Below are the different products based on the XML samples, please note some vendors have additional fields and advise reviewing the sample XMLs such as Finicity and Verify.

ID	EmploymentVerificationType	FulfillmentType	RecordFilter	Employer Info	Product	Vendor
01	Employment	Instant	Both	Optional*	TWNMortgageVOE	TWN
02	Employment	Instant	Select	Optional*	TWNEmploymentSelect	TWN
03	EmploymentPlusIncome	Instant	Both	Optional*	TWNMortgageVOE	TWN
04	Employment	Instant	MortgageSelectAll	Optional*	TWNMSAVOE	TWN
05	EmploymentPlusIncome	Instant	MortgageSelectAll	Optional*	TWNMSAVOI	TWN
06	EmploymentPlusIncome	Instant	TwelveMonth	Optional*	TWNEmploymentncome12Mo	TWN
07	EmploymentPlusIncome	Instant	TwentyFourMonth	Optional*	TWNEmploymentncome24Mo	TWN
08	EmploymentPlusIncome	Instant	Ultimate	Optional*	TWNMortgageUltimate	TWN
09	EmploymentPlusIncome	Instant	Preferred	Optional*	TWNMortgagePreferred	TWN
10	Employment	Manual	Specific	Required	ManualVOE	TWN, CRA
11	EmploymentPlusIncome	Manual	Specific	Required	ManualVOI	TWN, CRA
12	SelfEmployed	Manual	N/A	Required	Self-Employed3rdpartyPrep	TWN, CRA
13	SelfEmployed	Manual	N/A	Required	Self-EmployedSelfPrep	TWN, CRA
14	ReVerify	NA	N/A	N/A	ReVerify	TWN, CRA
15	VOIE	Instant	N/A	N/A	EmploymentIncomeConsumerDriven	Finicity
16	Employment	Instant	Current	N/A	VerifyCore	Verify
17	EmploymentPlusIncome	Instant	Current	N/A	VerifyPlus	Verify
18	EmploymentPlusIncome	Instant	Both	N/A	VerifyPremium	Verify
19	ReVerifyVOE	Instant	N/A	N/A	VerifyReverifyVOE	Verify
20	ReVerifyVOIE	Instant	N/A	N/A	VerifyReverifyVOIE	Verify
21	Employment	Instant	Current	N/A	VerifyCPDCore	Verify
22	EmploymentPlusIncome	Instant	Current	N/A	VerifyCPDPlus	Verify
23	EmploymentPlusIncome	Instant	Both	N/A	VerifyCPDPremium	Verify

24	ReVerifyVOE**	Instant	N/A	N/A	VerifyCPDReverifyVOE	Verify
25	ReVerifyVOIE**	Instant	N/A	N/A	VerifyCPDReverifyVOE	Verify

*Employer info should only be included under specific circumstances, which account for a small percentage of orders. End-users should be given the option to add employer info to these orders, but this should be an opt-in selection that they should have to make on each request. Including employer info where not required may return a NoHit response where it otherwise wouldn't.

**Only Verify CPD Core and Premium can be reverified.

The system will enforce the VOE cascade based on the customer setup. This helps ensure the proper flow of VOE vendors are ordered and suggest clients ordering outside of the website to be configured with the User Selected. That way clients can order the vendor they want rather than being restricted by a specific vendor order.

CONSUMER DRIVEN VERIFICATION OF INCOME & EMPLOYMENT

The Consumer-Driven verification operates differently than the other products. When an order is submitted, a URL is included with our initial response, which you will need to forward to the borrower. When the borrower navigates to the URL, they will be given an opportunity to upload paystubs and input banking information. VOIE orders can be processed if the applicant's employer reports information to automated services like Finicity.

MeridianLink does not store this URL. This means any follow-up polling requests will not contain the URL in the response. If a new order is detected to be a duplicate of a prior order, the response will include the details of the prior order, but the URL will still not be included. Users who need to resend the URL to the borrower may do so via our web interface.

APPENDIX

DUPLICATE ORDER DETECTION

SmartAPI utilizes a duplicate order detection mechanism to prevent accidental orders. If our system detects that an order is a duplicate of an earlier one, instead of creating a new file for your order and processing it, we will return the existing file we located. While this feature is helpful in a production environment, where duplicate orders unnecessarily increase costs, in a test environment, submitting duplicate orders is often intentional. To ensure consistent behavior between test and production environments, however, duplicate order detection is kept active on the test system.

Each product offered through SmartAPI utilizes its own logic regarding what it considers to be a duplicate and each product has its own lookback period (number of days it will look back for a duplicate order). Rather than outline the details of this logic for each product, it is simpler to describe what to look for to determine if a new order was determined to be a duplicate.

When a new order is submitted, SmartAPI will create a unique VendorOrderIdentifier and echo it in the response. In the scenario where a new order is submitted and the VendorOrderIdentifier returned is not unique (i.e. it matches something a prior order was already assigned), that means the request was caught by the duplicate detection logic. In such a case, the VendorOrderIdentifier echoed references the existing order and all information returned reflects that of the existing order.

Additionally, if you submit a new order and the StatusCode returned is Completed, the order was probably a duplicate and what you received was the prior order. New orders typically take some time to process, so if your new order completes quicker than you would normally experience, it was probably detected as a duplicate.

If you encounter difficulties during integration efforts as a result of duplicate detection, please consult with your technical contact for assistance with workarounds.

ERROR CHECKING

This section covers what to check for to see if errors were encountered in your request. There are 3 types of errors that our system may return in the response XML.

REQUEST-LEVEL ERROR

If this error appears, it means the request failed very early in the process. This type of error can be triggered by submitting a request that does not pass schema validation (e.g. mis-keying an element name) or by submitting invalid login credentials. Notice that this type of error isn't specific to the product (i.e. SERVICE) being requested, but rather with the request itself. If the following node exists, that means a request-level error was encountered:

/MESSAGE/DEAL_SETS/DEAL_SET_SERVICES/DEAL_SET_SERVICE/ERRORS

Within this node, there exists the error code and the error text, as provided below. These values should be extracted and displayed to the user along with an indication that an error was encountered.

/MESSAGE/DEAL_SETS/DEAL_SET_SERVICES/DEAL_SET_SERVICE/ERRORS/ERROR/ERROR_MESSAGES/ERROR_MESSAGE/ErrorMessageCategoryCode/text()

/MESSAGE/DEAL_SETS/DEAL_SET_SERVICES/DEAL_SET_SERVICE/ERRORS/ERROR/ERROR_MESSAGES/ERROR_MESSAGE/ErrorMessageText/text()

PRODUCT-LEVEL ERROR

The product-level error is the error type that most of the integration guide refers to in the workflow sections. The presence of this error indicates the service itself encountered an issue. For example, failing to include an SSN when ordering a Consumer Credit report or attempting to order a VOE when your account isn't enabled by the service provider for this product.

Product-level errors can be identified when the service's StatusCode node contains a text value of 'Error'. XPath below:

//SERVICE/STATUSES/STATUS/StatusCode[text()='Error']

When that is the case, a description of the error may also appear. This description should be extracted and shown to the user, along with an indication that an error was encountered.

//SERVICE/STATUSES/STATUS/StatusDescription/text()

To simulate a product-level error, try submitting a consumer credit report order where the SSN is only 8 digits, instead of the standard 9.

FORMAT ERRORS

Format errors occur when issues are identified with the data in your request, but aren't significant enough to stop us from processing the request. These are more like warnings than they are errors, as they are for informational purposes. An example of a format error would be ordering consumer credit where the borrower's middle name contains numbers. We will echo back a format error in the response, but your request will still process because the middle name is not a required field. In such an event, we would just skip the middle name and process everything else.

Format errors are specific to the service itself. It will appear, if applicable, under the SERVICE container. The node to look for is the following:

//SERVICE/ERRORS

Format errors won't have a code, but will always have a description:

//SERVICE/ERRORS/ERROR/ERROR_MESSAGES/ERROR_MESSAGE/ErrorMessageText/text()

Format errors do not need to be shown to the user. If you prefer to show these to the user, we suggest conveying them as "tips" or "advice" and not as "errors encountered". Regardless, you should log these errors somewhere and review occasionally to see where adjustments to your system can be made.

ENUMERATIONS

NAME SUFFIX

The following are valid enumerations for the consumer name suffix:

SR, JR, I, II, III, IV, V, VI, VII, VIII, IX

INTERFACETOKEN

An HTTP header of “InterfaceToken” can be included with your XML request if you have a value or values that you’d like echoed in our immediate response. This can be a tracking number, a comma-separated list of values, etc. We recommend encoding the value in Base64 to minimize unintended effects of any special characters that may be present in your value, though this is not required.

NAMESPACES & SCHEMAS

NAMESPACE PREFIXES

This guide contains many XPaths, referencing nodes used in the request packet and in the response packet. Any prefixes that appear within these XPaths can be referenced here.

REQUEST XML

Prefix	URI
(Default)	http://www.mismo.org/residential/2009/schemas
p2	http://www.w3.org/1999/xlink
p3	inetapi/MISMO3_4_MCL_Extension.xsd

RESPONSE XML

Prefix	URI
(Default)	http://www.mismo.org/residential/2009/schemas
d7p1 or d5p1	http://www.w3.org/1999/xlink
p3	inetapi/MISMO3_4_MCL_Extension.xsd
xsd	http://www.w3.org/2001/XMLSchema
xsi	http://www.w3.org/2001/XMLSchema-instance

XML SCHEMAS

Schemas are located under the MISMO Resources folder in the Smart API Integration Kit. There are 5 different schemas.

MISMO 3.4 EXTENDED:

This is the condensed and extended version of the original MISMO schema. The types of unused container elements from original MISMO schema have been replaced to UNUSED_CE_TYPE and the types of unused data points have been replaced to Unused_SE_Type. Attempts to validate your request XML documents should be ran against this file.

MISMO 3.4 MCL EXTENSION

This schema contains extension types to MISMO schema and it is imported by the MISMO 3.4 Extended schema.

XLINK MISMO B324

This schema contains XLink definitions and is imported by the MISMO 3.4 Extended schema. It has been taken from MISMO schemas with the addition of MCL defined Arc Roles (i.e. MCLServiceArcroles and MCLDealArcroles).

XLINK

This schema contains XLink definitions and included by the XLink MISMO 3.4 schema. It has been taken from MISMO schemas without any changes.

XML

This schema contains a few xml definitions and imported by the XLink schema. It has been taken from MISMO schemas without any changes.

RESPONSE FORMATS

Up to 3 different response formats are available for each service. Please refer to each product's corresponding XML samples to understand how response formats are requested.

- PDF – The completed report will be available for viewing as a PDF document. The document is provided as a base64-encoded string within the response.
- HTML –The completed report will be available for viewing as a HTML document. The document is stored within a CDATA container within the response.
- Parsable XML – While the response itself is an XML document, what Parsable XML means is that various data from the completed report are included as XML nodes, which can then be extracted via XPath. For example, the parsable XML data included with the Consumer Credit Report would be information about liabilities, public records, scores, etc.

The table below shows which response formats are supported by each product. Requesting a format not supported by a product will result in an error response.

Service	PDF	HTML	Parsable XML
Automated Valuation Model (AVM)	X	X	X
Business Credit Report (Experian)		X	
Business Credit Report (Equifax)	X		X
Consumer Credit Report	X	X	X
Criminal Record Report	X		X
Eviction Report	X		X
Flood Zone Determination	X		X*
InstantID	X	X	X
Mortgage Electronic Registration System (MERS)	X	X	X
Mortgage Participant Verification (MPV)	X	X	X
Packaged Report	X	X	X
Precise ID Knowledge IQ	X	X	X
Precise ID Score	X	X	X
SSA-89	X		X
Tax Transcript (TAX)	X		X*
Undisclosed Debt Notification (UDN)	X	X	X
Verification of Asset (VOA)	X		X
Verification of Deposit (VOD)	X		X
Verification of Employment (VOE)	X	X	X
VOIE Consumer Driven	X		X

*Only supported with certain VSPs/vendors. Speak to your technical contact if you have questions.

SURROGATE LOGIN

Surrogate Login is a feature that allows a user to access the service provider's system and impersonate another user. Once impersonation begins, any privileges or restrictions are based off the account being impersonated. Depending on how your loan software functions and your typical end-user's access permissions, surrogate login is something you may or may not need to implement. Below, we will cover a typical scenario where surrogate login allows users a degree of flexibility, while maintaining account security.

Sara, the Office Manager, and Jim, a Broker, both work at ABC Mortgage. Jim is working from outside the office today and meets a potential client. This client is actively searching for a broker, but doesn't know if his credit score meets the minimum requirements for a loan. Jim knows if he cannot give this person an answer before they part ways, he will likely lose this opportunity. Jim calls the office and speaks with Sara. He informs her of his need to order a credit report for the potential client. Sara understands and accesses the service provider's system.

With their previous service provider, Jim would normally give Sara his login and password, so that she could log in and order the credit report under his account. However, ever since they've started using a service provider that supports surrogate login, Sara is able to access using her own login credentials and instruct the system that she wants to impersonate Jim. When the credit report is ordered, it's as if Jim were the one who ordered it. Sara relays the credit scores to Jim over the phone and all is well. Ownership of the file is automatically assigned to Jim, along with any associated charges. The audit logs will also confirm that Sara logged in and impersonated Jim, in case there is any future dispute. Most importantly, Jim did not have to disclose his login and password to Sara, so they did not violate their company's security policy.

As we can see, the scenario above makes great use of surrogate login, but not every company needs this feature. Companies where the concept of file ownership is non-existent or where the company pays for all products ordered may not have a need to implement this feature.

When discussing surrogate login, the following two terms will be used to identify the parties involved:
Authenticating User: The person who is logging in. This person provides their login and password to gain access to the system.
Surrogate User: This is the person being impersonated. Only the login name of the surrogate user is required to indicate who the authenticating user wants to impersonate. At no time is the surrogate user's password needed.

The below list outlines important things to be aware of with regards to this feature:

- To utilize surrogate login, an additional header of 'MCL-SurrogatedLogin' should be sent with the HTTP request. The value for this entry should be the login name of the surrogate user, exactly as it appears. Unlike the 'Authorization' header, the 'MCL-SurrogatedLogin' value must be sent in plain text, not base64.
- If there are any login restrictions applied to the surrogate user's account, they will be ignored. This is because the surrogate user is not actually logging into the system: the authenticating user is. With that said, any login restrictions that exist on the authenticating user's account will remain in place.
- File access or product ordering permissions will be based off the configuration of the surrogate login. For example, if Sara is allowed to order credit reports, but Jim is not, then any attempts by Sara to order credit reports for Jim by impersonating him will fail. This is because Sara's account is only doing the authenticating. Jim's account is the one being impersonated, so any permission (or restrictions) on his account remain in effect.
- There are limits on who the authenticating user can impersonate. For example, the authenticating user will not be able to impersonate an employee of an unrelated company that uses the same service provider. Similarly, if the authenticating user is not given permission to impersonate other users, they will not be able to. These types of details are typically handled by the service provider and end-user administrators. As far as integration goes, the software developer need only ensure that any error messages returned by our system are displayed to the end-user. Such error messages will let the user know if there is an administrative restriction preventing the surrogate login attempt from being successful.

If you would like to implement surrogate login, please speak with your assigned technical contact.

TEST CASES

Below are test cases that can be used with SmartAPI. By default, test accounts are only enabled for Consumer Credit Report ordering. If you need to test other services, please speak with your MeridianLink technical representative to ensure your account is properly configured. The following notice applies:

- The MCL test system should only be used by personnel involved with integration efforts. End-users of your software should not be given unmonitored access to submit orders on this system.
- Some vendors to which MCL integrates may return real data. Please do not abuse the test system. MeridianLink has the right to terminate your access to the test platform if we detect abuse.
- Please avoid running long-term automated requests against the test system. If you have requirements that can be accomplished by doing this, please discuss with your MCL technical representative.
- Test cases should only be used in the testing environment unless otherwise noted.

AUTOMATED VALUATION MODEL (AVM)

Any address can be used for testing; live data may be returned. Addresses known to return results with most valuation models are listed below.

Address	APN
6788 Moselle Dr, San Jose, CA 95119	70439023
2178 Toryglen Way, San Jose, CA 95121	67647025
1587 Princeton Drive, San Jose, CA 95118	56732002
1024 Jenifer St, Madison, WI 53703	070913402221
569 Queen St, Belleville, WI 53508	050834230836
219 Schley Pass, Madison, WI 53703	071007226379
2020 Silver Ave, Las Vegas, NV 89102	16204210063
1229 Park Circle, Las Vegas, NV 89102	16204114017
4435 Sirius Ave, Las Vegas, NV 89102	16207411064

BUSINESS CREDIT REPORT

Any business can be used for testing; live data may be returned. The below test cases are known to be reliable in returning data.

Name	Address	EIN	Phone	Notes
EXPERIAN TESTCASES				
Hong Kong Express	6767 W Sunset Blvd, Los Angeles, CA 90028			Standard Test
IBM Corp	1 New Orchard Rd, Armonk, NY 10504	130871985	914-499-1900	Standard Test
Microsoft Corp	1 Microsoft Way, Redmond, WA 98052			Standard Test
Walmart Inc.	702 SW 8th St, Bentonville, AR 72716	710415188	479-273-4000	Standard Test
ABC Mortgage	123 Main St, Santa Ana, CA 92701			No Hit
Experian Information Solutions	123 Main St, Costa Mesa, CA 92626			No Hit, Similars Found
Walmart Inc.	14081 Magnolia St, Westminster, CA 92683			No Hit, Similars Found

EQUIFAX TESTCASES			
10 PERCENT PRODUCTIONS	6165 SANTA MONICA BLVD, LOS ANGELES, CA 90038	323-460-4661	Standard Test
100 MT. CARMEL ROAD, LLC	13038 BEAVER DAM RD, COCKEYSVILLE, MD 21030	410-252-8361	Error – No Matches
1 3 SUN, INC	6770 FRANKLIN BLVD STE 2, SACRAMENTO, CA 95823	916-421-9473	Standard Test

CONSUMER CREDIT REPORT

The below test cases can be used for consumer credit ordering. Note that entries with last name 'Testcase' will return static score and fraud information. If your platform will include the corresponding flags to indicate score or fraud add-on preferences, we recommend testing with entries that do not have a last name of 'Testcase'.

Name	SSN	Address	Notes
Brett K Ingram	666-68-3160	5810 County Line Rd, Indianapolis, IN 46235	Equifax: Frozen File Transunion: Frozen File
Joshua Tree	666-43-4321	861 Desert T, Palm Springs, CA 92262	Transunion: Frozen File
Margaret F Rrtzx	666-78-4293	5178 Nubsztyllq Pl, Frtw, IN 46804	Equifax: Frozen File
Ann Gage	000-00-0003	123 Rocky Rd, Anthill, MO 65488	Experian: Frozen File
Anabel Farfan Sr.	666-46-3461	27484 Acacia Dr, Valencia, CA 91354	Equifax: No Record Found Experian: No Record Found
Abcd Efghi	666-12-1234	123 Main St, Santa Ana , CA 92705	Equifax: No Record Found Experian: No Record Found TransUnion : No Record Found
Douglas D Abt	666-97-6523	1601 N Hollywood Way Unit K, Burbank, CA 91505	Equifax: No Record Found Experian: No Record Found TransUnion: No Score, Lack of Credit
Patricia M Klkgk	666-46-3533	225 I St SE, Auburn, WA 98002	Equifax: No Score, Deceased Experian: No Record Found TransUnion: No Record Found
Sheila Wild	666-11-0070	3042 N County Road 69, Manila, AR 72442	Equifax: No Score, Lack of Credit TransUnion: No Record Found
Charles J Hhoverton	899-48-6816	PSC 808 BOX 74, DPO, AA 34008	Military Address Example
Edward A Eeagle	897-89-4652	CMR 5775, DPO, AE 09742	Military Address Example
Jose Martinez	666-64-5703	575 Main St, Chatham, NJ 07928	OFAC Hit
Joseph E Gilbert	666-44-0268	6659 Washington, St. Louis, MO 63130	OFAC Hit
Muhammad M Abdullah	666-12-1212	80 Ritz Cove, Dana Point, CA 92629	OFAC Clear
Karen Aahtexo	666-79-3551	899 Wyxbkhw Rd, Salem, VA 24153	Standard 3-bureau test consumer.
Carl Ttjhtkh Sr	666-79-4386	7955 Cdpwkydj St Apt 2, Waco, TX 76705	Standard 3-bureau test consumer.
John Ppklx	666-79-3852	1596 Twxduxy Ln, Oceanside, CA 92056	Standard 3-bureau test consumer.

Janice Llwwt	666-79-5316	739 Excd Mjyhxt Rd, Jackson, GA 30233	Standard 3-bureau test consumer.
Joyce Becraft	666-41-8776	628 County Rd 503, Hanceville, AL 35077	Standard 3-bureau test consumer.
Diane Bruso	666-94-6911	5968 Georgia, Birmingham, AL 35212	Standard 3-bureau test consumer.
Jon Balloch	745-88-8966	1 Johnson St, Dauphin, MB R7N2L4, CA	Canadian Consumer.
James Adam	000-00-0000	206 Simpson, Bowmanville, ON L1C2J1, CA	Canadian Consumer.
Bobby Test	000-00-0000	833 John St, Hamilton, ON L8N1L4, CA	Canadian Consumer.
The below testcases are exceptions and can be used in both the testing and production environment			
*Marisol L Testcase	000-00-0001	220 Locust Ave, Anthill, MO 65488	~740 Score
*David C Testcase	000-00-0002	504 Grandview St Unit 2, Anthill, MO 65488	~660 Score
*Janet X Testcase	000-00-0003	19 Forest Dr, Anthill, MO 65488	~620-670 Score
*Joseph M Testcase	000-00-0004	13743 Bluewater Cir, Anthill, MO 65488	~620 Score
*Maria A Testcase	000-00-0005	5524 Eastbourne Dr, Anthill, MO 65488	~520-570 Score
*Carlos E Testcase	000-00-0006	9614 Oriole Ln, Anthill, MO 65488	~490-530 Score
*Nicole S Testcase	000-00-0007	8315 Bunker Hill Rd, Anthill, MO 65488	~500-530 Score
*Sarah D Testcase	000-00-0008	171 Glenwood Dr, Anthill, MO 65488	<500 Score
*Luis T Testcase	000-00-0009	5002 Banner Ct, Anthill, MO 65488	<500 Score
*Bill C Testcase	000-00-0015	8842 48th Ave, Anthill, MO 65488	Co-Borrower: Beth C Testcase 000-00-0016
*Condi X Testcase	000-00-0017	784 Beach Blvd, Anthill, MO 65488	Co-Borrower: Cary X Testcase 000-00-0018

CRIMINAL RECORD REPORT

Below are test cases that can be used for the Criminal Records Report. Address is required, but any address can be used.

First	Last	SSN	DOB
Darryl	Antarr	999-99-9999	02/28/1965
David	Morgan	999-99-9999	12/22/1961
Jonathan	Consumer	999-99-9998	01/01/1971

EVICTION REPORT

Below are test cases that can be used for the Eviction Report Product. SSN and address are required, but any can be used.

First	Last	Test-Case Scenario
Jonathan	Consumer	Single Hit-National
Kris	Consumer	Single Hit-Western
Indiana	Consumer	Two Hits-One Western, One National
Holly	Consumer	Four Hits
Erika	Consumer	No Hits

Adam	Consumer	Notice-National-One InProgress
Bobbie	Consumer	Notice-National-One Invalid SSN
Christina	Consumer	Notice-National-One Service Unavailable
Darren	Consumer	Notice-National-One Communication Error
George	Consumer	Message-Disputed Applicant

FLOOD ZONE DETERMINATION

Any address can be used for testing; live data may be returned. The below test cases are reliable in returning the indicated result. Most orders will complete after a brief period; a small number will require manual review. You can contact your MeridianLink technical provider to set these to completed or you can leave them in their current state indefinitely. We recommend assigning a unique LoanIdentifier value to each test submission.

Address	Result
800 Lakeshore Pkwy, New Orleans, LA 70124	Success
180 Troon Trail, Durango, CO 81301	Success
412 All Saints Pl, Raleigh, NC 27607	Success
10480 Borror Rd, Orient, OH 43146	Success
587 Hummingbird Ln, Fredericksburg, TX 78624	Success
777 Hummingbird Ln, Fredericksburg, TX 78624	Manual
139 Cypress Ct, Durango, TX 78256	Manual
1433703 Tullis Dr, New Orleans, LA 70131	Manual

INSTANTID

Below are test cases for the InstantID product.

Name	SSN	Address	DOB	Home Phone
Ronald E Sanchez	334-88-6975	20004 Eros Dr., Olympia Fields, IL 60461	01-19-1967	864-254-8568
Laura Carreno	332-82-0235	514 Palmyra St, Dixon, IL 61021	03-28-1980	678-850-0561
Brandon Haroldson	555-61-1303	1433 N Linder Ave, Chicago, IL 60651	09-07-1977	803-298-0961
Willie Ebrahimi	617-86-6348	2910 S Dearborn St Apt 30, Chicago, IL 60616	12-28-1978	803-460-7096
Fermin Townsend	572-81-1689	2105 Bonnycastle Ln, Yukon, OK 73099	05-27-1970	
Stacey Hanerfeld	563-81-8817	465 W Gonzales Rd Unit 20, Oxnard, CA 93036	06-18-1981	813-422-2934
Felix Levens	611-20-8737	1609 Line St Apt A5, Decatur, GA 30032	10-18-1946	336-632-1558
Victoria Velazquez-Aguiar	565-53-6396	546 Nebraska Ave, Pontiac, MI 48341	04-01-1986	
Chelsea Botello	555-83-9903	16150 Gilchrist St, Detroit, MI 48235	02-17-1960	805-229-9325
Timothy Bowman	618-12-1451	10191 Pasadena, Ferndale, MI 48220	10-04-1992	626-392-9542

MORTGAGE ELECTRONIC REGISTRATION SYSTEM (MERS)

Below are test consumers that will return results for the MERS service.

First Name	Last Name	SSN
Jane	Consumer	000-00-0001
Mort	Gage	000-00-0002
Sam	Tester	000-00-0003
Ken	Customer	500-50-7000
John	Homeowner	999-40-5000

Below are test subject properties that will return results.

Subject Property Address
2122 Trinity Ave, Mojave, CA 93501
123 Main St, San Diego, CA 92128
145 New Jersey #10, Whippany, NJ 07981
12345 Candy Cane Ln, Costa Mesa, CA 92626
10655 Birch St, Burbank, CA 91502

MORTGAGE PARTICIPANT VERIFICATION (MPV)

Any information can be used for testing; live data may be returned. The below test cases are known to be reliable in returning data.

Participant Type	Name	Role	License #	State
Individual	Valerie Brown	Appraiser	A12348688	MA
Individual	John Snowe	Appraiser		AL
Individual	Michael Burns	Trustee		TX
Individual	John P Brown	MortgageBroker		FL
Business	Summit Advisory Partners	Investor		MN
Business	GL Systems LLC	Investor		DE
Business	Acceleration Capital, LLC	Investor		CA
Business	Advanced Trading Workshop	Investor		NY

PACKAGED REPORT

For test borrowers, see Consumer Credit Report test cases. For test subject properties, see the MERS test cases. For test participants, see the MPV test cases.

PRECISE ID KNOWLEDGE IQ

Below are test consumers that can be used for the Precise ID KIQ product. The out-of-wallet questions and corresponding answers may change over time, so we have decided to not document them here. To test this product, you will need to coordinate with your technical representative to obtain the answers to the questions that are presented with each order.

Name	SSN	DOB	Address
Victoria E Austin	666-78-5742		53555 County Rd, Holyoke, CO 80734

Laurieann Kathleen Radlein	666-13-5502	02-14-1968	3507 Woodlea Ave, Baltimore, MO 26214
Gordon G James	666-04-5883	03-14-1963	603 W El Norte Ave, Monrovia, CA 91016
Julie A Caputo	666-72-4863	01-20-1959	1903 Ingersol Pl, New Port Richey, FL 33552
Timothy F Wiklund	666-58-1819		4711 Boulder Hwy #F1, Las Vegas, NV 89121
Helen Mulligan	666-94-4614	05-23-1960	22 Park Ave, Portland, ME 04101

PRECISE ID SCORE

Below are test cases that can be used for the Precise ID Score product.

Name	SSN	Address	DOB and Phone	DL# and State
Mathew J Doyle	666-36-1452	10046 Colonnade Dr., Tampa, FL 33647	09-02-1933	D452667333220 State: FL
Preston Oen	666-17-4540	3685 Yellowstone Ave., Dayton, OH 45416	11-21-1968	RS447974 State: OH
Julie A Caputo	666-72-4863	1903 Ingersol Pl., New Port Richey, FL 33552	01-20-1959 561-736-6450	C140561595200 State: FL
Belinda Carter	666-05-6157	1456 Ivory Dr., Santa Maria, CA 93455	04-15-2000	
William N Johnson	666-23-5464	2505 Laney Cir., Denton, TX 76208	01-01-1970	15577427 State: TX
Gary Wilton	666-64-9524	212 Patton Dr., Pearl, MS 39208	06-17-1947 601-420-4323	4963927 State: AL

SSA-89

The SSA-89 product takes a few minutes to process and normally requires no manual intervention. However, depending on your VSP's agreement with the end-user, the VSP may place all orders in a hold status for manual review before releasing it for processing. Please contact your MeridianLink technical provider to coordinate testing the various scenarios for this product.

Name	SSN	DOB	SSA Results
John Sample	223-34-4555	01-01-1980	SSN is valid
John Test	223-34-4557	01-01-1980	SSN is invalid
John Businessman	223-34-4556	01-01-1980	SSN holder is deceased

TAX TRANSCRIPT (TAX)

Testing the TAX product will require assistance from your MeridianLink technical contact. Any information may be used for the test case. Your technical contact will manually process these orders.

UNDISCLOSED DEBT NOTIFICATION (UDN)

The test environment for the UDN product is pending further development. You may use any consumer information for your test orders, but the orders will not cycle through their statuses automatically. To perform comprehensive testing of the UDN product, please coordinate with your MeridianLink technical representative.

VERIFICATION OF ASSET (VOA)

Please coordinate with your MeridianLink technical contact to handle testing for this product. Your technical contact will act as the borrower. When submitting a VOA order, you will want to set the borrower's email field to that of your technical contact's.

VERIFICATION OF DEPOSIT (VOD)

Deposit verifications are manually processed by your VSP. Because these are a manual process, please coordinate with your MeridianLink technical contact to test these. Your technical contact will act as the VSP and process your orders.

VERIFICATION OF EMPLOYMENT (VOE)

The below test consumers can be used with instant lookup orders. At the time of this writing, the indicated results are only reliable when submitting an EmploymentVerificationType of Employment with no employer specified. If testing inclusion of the optional employer name, please pull a test case that will return results, then view the report to see what employers the test consumer has on record and resubmit the order with that employer's name specified. For orders being processed manually, your technical representative will act as the service provider and can process these at your request.

SSN	Notes
001-01-2401	No Hit.
001-01-2402	No Hit.
001-01-2403	No Hit.
001-01-2404	Current employment records found. Order will complete.
001-01-2405	Current Employment records and Previous employment record found. Order will complete.
001-01-2406	Current Employment records and Previous employment record found. Order will complete.
001-01-2407	Current employment records found. Order will complete.
001-01-2410	Returns an error of "Multiple individuals associated with this SSN."
333-33-3333	Returns an error of "Multiple individuals associated with this SSN."
666-13-0418	Returns an error of "Employer requires a valid salary key."
799-00-0002	Previous employment records found. Order will complete.
799-00-0003	Current and Previous employment records found. Order will complete.
799-00-0004	Current and Previous employment records found. Order will complete.
799-00-0011	Current and Previous employment records found. Order will complete.
799-00-5001	Current employment records found. Order will complete.
799-00-5002	Current Employment record and Previous employment records found. Order will complete.
799-00-5003	Current and Previous employment records found. Order will complete.
799-00-5004	Returns an error of "Multiple individuals associated with this SSN."
799-00-5005	Current employment records found. Order will complete.
799-00-5141	This is one where employer name needs to be provided, along with accurate applicant name for a successful result. Use Employer Name of "Enterprise One" (or Employer Code of 91001) and borrower name of Test RhodeIslandEmpl.
799-00-5142	This is one where employer name needs to be provided, along with accurate applicant name for a successful result. Use Employer Name of "Enterprise One" (or Employer Code of 91001) and borrower name of Test EmplRhdIslnd.

999-00-4108	Current employment records found. Order will complete.